

Source: NASA



The backshell as captured by the Ingenuity helicopter

not stop there, as the copter set a distance record a couple of days later. Come April 25, the team controlling the helicopter from NASA's Jet Propulsion Laboratory in Southern California pushed it to fly and cover a horizontal distance of 100 metres. The altitude gained remained the same as the last flight, but the time shot close to a minute and a half, approximately 80 seconds.

These results were more than what anyone expected to get in the first place. During that and the subsequent flights, the helicopter has managed to capture pictures of its own while in the air. NASA has made these images captured by Ingenuity and other interesting flight-related information public, too, which can be accessed by visiting the following URL: <https://digit.in/marscopter>.

As the helicopter had shown its capabilities, demonstrating its controlled and powered flight capabilities on Mars, the researchers in charge of the 'experiment' set out to explore further opportunities as they assigned a new mission to the helicopter. Ingenuity was moved to the operations demonstration phase, aimed at exploring further possibilities of using the aerial scouting capabilities

of the aircraft. This phase was aimed not only to further the exploration possibilities of Mars but other planets in the coming future too. The results from this phase are also expected to open doors to the exploration of Mars by astronauts using helicopters for navigation on the Red Planet. The images captured by the helicopter are being put to use by the researchers at NASA to support the Perseverance Rover's exploration of the Jezero Crater.

In its 10th flight on the planet, on July 24, 2021, the helicopter completed its first mile (1.6 km) of flight on Mars, by covering about 764 feet (233 metres) of horizontal distance, in its highest flight thus far, where it climbed to a height of 12 metres. The helicopter travelled its longest distance when it covered 704 metres in 161.3 seconds on the April 8, 2022.

The most recent example of the image capturing and exploratory activities of the Ingenuity helicopter are the pictures of the backshell and supersonic parachute that helped the Perseverance rover land on the surface of Mars. These images, which were beamed back to Earth, were captured on the same day, a year later from the helicopter's very first flight on the April 19 (More details on this later).

The power for the helicopter is harnessed from the rays of the Sun. Solar power captured using solar panels is used to charge the batteries of the helicopter. Standing at a height of about 19 inches and with a weight of 1.8 kgs, the Ingenuity helicopter is fitted with counter-rotating blades which are capable of rotating at about 2,400 rpm. It has two cameras and computational systems, which help it not only operate autonomously in certain conditions but also process and relay the data it gathers.

A year after the helicopter rose from the surface of Mars for the first time, on the April 19, 2022, it completed a flight which lasted for 159 seconds. While this is not its longest flight in terms of time duration ever, it captured images which showed the other side of the Mars landings of 2020, which were dubbed to be one of the most stressful moments for the team handling the entire mission. The images captured begin from the moment the helicopter lifts off from the surface of the planet to the time it completes its flight path, hovering over the debris of the Perseverance landing, and is about to land on the surface of Mars.

For anyone connected to the world of space science and technology, these strides taken by the helicopter are bound to make them even more keen in looking for the future prospects on the planet that has for so long been touted to be the place where humans might settle next. Coming to the regular folks, there's also something on the plate. These developments are ones to be kept an attentive eye on for anyone slightly interested in the prospect of taking a spacecraft to Mars when some space agency, either government or private, eventually finds a way to raise a human settlement there. This is because when on Mars, you would rather prefer taking a flight from one place to another rather than driving on the planet's dusty roads, and for that to happen, this helicopter would play an important role. d